Course Handout

Institute/School Name	Chitkara University Institute of Engineering & Technology		
Department Name	Department of Computer Science and Engineering		
Programme Name	B.E - Computer Science & Engineering		
Course Name	Backend Engineering-I	Session	Jan – June 2026
Course Code	24CSE0214	Semester/Batch	4th /2024
L-T-P(Per Week)	2-0-2	Course Credits	3
Pre-requisite	Basic Knowledge Java Script and React	NHEQF Level[1]	5
Course Coordinator	Dr. Kiran Deep Singh	SDG Number[4]	1,2,9

1. Objectives of the Course

The scope of the course is to Gain a comprehensive understanding of backend development concepts, including server-side languages, databases, APIs, and server management.

- Students will master one or more programming, such as Javascript and Node.js, focusing on their application in server-side operations.
- Develop skills in designing and implementing databases.
- Learn how to design, build, and test RESTful APIs to facilitate communication between front-end and back- end systems.

2. Course Learning Outcomes (CLOs)

Student should be able to:

	CLOs	Program Outcomes (PO)	NHEQF Level Descriptor[2]	No. of Lectures
CLO01	Understand the fundamental concept of client-server architecture and implement server-side logic.	PO3, PO4, PO10,PO11, PSO1, PSO2	Q1, Q2, Q3	**8**
CLO02	Build web applications using express.js, including routing, middleware integration, and response handling.	PO3, PO4,PSO1, PSO2	Q2, Q3	**8**
CLO03	Apply Blocking vs non-blocking code -Body parser	PO5, PO7, PO11, PSO1, PSO2	Q2, Q3, Q4	**14**
CLO04	Analyse the EJS templates to create dynamic and reusable front-end components.	PO11, PSO1, PSO2	Q3, Q4, Q5	**10**
CLO05	Design the minor project and generate report using backend Technologies.	PO7, PO9,PSO1, PSO2	Q4, Q5	**10**
CLO06 (Only for lab components)	Demonstrate Backend engineering skills through end-to-end project development, integration, and deployment.	PO5, PO10, PO11, PSO1, PSO2	Q3, Q4, Q5, Q6	**10**
Total Contact Hours				**60**

CLO-PO Mapping

CLO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	Type of Assessment's[3]
CLO01			M	M						H	H	H	M	Formative
CLO02			M	M								H	M	Formative
CLO03					H		M				M	H	M	Formative
CLO04											H	M	H	Formative
CLO05							M		H			M	H	Formative
CLO06					M					M	H	M	H	Formative

H=High, M=Medium, L=Low

3. Recommended Books:

B01: Node.js, MongoDB and Angular Web Development, 2nd Edition, Brad Dayley, Brendan Dayley, Caleb Dayley, Oreilly Publications, 2017.
B02: Beginning Node.js, Express & MongoDB Development by Greg Lim, September 2020
B03: Node.js Design Patterns: Design and Implement Production-Grade Node.js Applications Using Proven

[1] National Higher Education Qualification Framework Level, Refer to annexure
[2] NHEQF Level Descriptor, Refer to Annexure & Learning outcomes descriptors for qualification for all levels on the NHEQF
[3] Types of Assessments can be referred from Type of Assessments. Refer to Annexure.
[4] For SDG Mapping with Courses, Pl refer SDG Mapping policy for Courses

Patterns and Techniques, Mario Casciaro and Luciano Mammino, Packt Publishing , 3rd Edition (2020)

Reference Books:

B04: Web Development with Node and Express: Leveraging the JavaScript Stack, Ethan Brown O'Reilly Media, 2019

B05: SQL Database Programming Hardcover by Tim Warren, 2020

E-Resources:

https://library.chitkara.edu.in/subscribed-books.php

4. Other readings and relevant websites:

SerialNo	Link of Journals, Magazines, websites and Research Papers
1.	https://nodejs.org (Docs + Examples)
2.	https://www.npmjs.com (Docs + Examples)
3.	https://firebase.google.com/docs/server/setup
4.	https://www.mongodb.com/products/platform/atlas-database

5. Recommended Tools and Platforms

- Node JS, React JS
- NPM, VS Code, MongoDB Atlas.

6. Course Plan: Theory+ Lab Plan

Theory Plan

Lecture Number	Topics
1-4	Detail Discussion of Course Handout (CHO) Introduction to Client Server Architecture, Understanding Server and how request is handled at the server.
5-8	Installing NodeJS and setup the environment, file handling module and diving into file dependency.
9-12	Understanding Nodejs, advantages and disadvantages with respect to the available languages.
13-16	Handling request, creating endpoints using HTTP module. Modules, NPM, import Modules.
17-20	Frameworks, Different frameworks of NodeJS starting with Express-JS.
21-24	Serve static files, Routing (Routing methods, Route paths, Route parameters, Route Handlers), Response Methods. Handling static pages with file stream, Handling exceptions.
	Project Based Evaluation-I (Lecture number 1-24)
25-28	Middleware: Middleware lifecycle, Application-level, middleware, Router-level Middleware, Error-handling middleware and Third-party middleware, how request travels in express, Blocking vs non-blocking code -Body parser.
29-32	SSR vs CSR and Diving into Template Engines, how to use the EJS/HBS template in your Node.js application
33-36	Introduction to databases, SQL and No SQL Databases, Connecting MongoDB, Learning ODM-Mongoose with NodeJS application.
37-40	Diving into session management and understanding Cookies and Express-Sessions.
41-44	Diving into authentication in Express-JS with Bcrypt, and understanding how JWT token works, and passport JS.
45-48	Deep Diving into full duplex communication and mastering socket.io
	Project Based Evaluation-II (Lecture number 1-48)
49-52	Starting with PostgreSQL CRUD operation, and deep diving into PRISMA.
53-56	Understanding Multer, Cloudinary to handle form-data.
57-60	Learning unit testing and deployment on Vercel/Render/AWS etc.
	Final Project Based Evaluation-III

Lab Plan

S. No.	Experiments	Learning Resource
1	Testpad Module Progress and Completion	Student's Testpad Portal of Chitkara University

7. Delivery/Instructional Resources Theory/lab Plan:

Lecture No.	Experiment / Topics	CLO	TLM	ALM	Web References	Audio-Video
1–4	CHO discussion, Client–Server Architecture, request handling	CLO-1	PPT, Board	Discussion	https://www.geeksforgeeks.org/system-design/client-server-architecture-system-design/	https://www.youtube.com/watch?v=a5CgfS0Y4Uc&pp=ygUuQ2xpZW504oCTU2VydmVyIEFyY2hpdGVjdHVyZSwgcmVxdWVzdCBoYW5kbGluZw%3D%3D
5–8	NodeJS installation, environment setup, file handling	CLO-1	Lab PC	Hands-on	https://www.geeksforgeeks.org/node-js/how-to-set-up-your-node-js-and-express-development-environment/	https://www.youtube.com/watch?v=OFbSqd54Wwk&pp=ygU1Tm9kZUpTIGluc3RhbGxhdGlvbiwgZW52aXJvbm1lbnQgc2V0dXAsIGZpbGUgaGFuZGxpbmc%3D
9–12	NodeJS concepts, pros & cons	CLO-2	PPT	Discussion	https://www.geeksforgeeks.org/node-js/the-pros-and-cons-of-node-js-in-web-development/	https://www.youtube.com/watch?v=q-xS25lsN3I&pp=ygUcTm9kZUpTIGNvbmNlcHRzLCBwcm9zICYgY29ucw%3D%3D
13–16	HTTP module, endpoints, npm, modules	CLO-2	Editor	Practical	https://www.w3schools.com/nodejs/nodejs_http.asp	https://www.youtube.com/watch?v=ZQsrcayZcSk&pp=ygUkSFRUUCBtb2R1bGUsIGVuZHBvaW50cywgbnBtLCBtb2R1bGVz
17–20	NodeJS frameworks, ExpressJS	CLO-3	Editor	Demo	expressjs.com	https://www.youtube.com/watch?v=pKJ4GGyDgJo&pp=ygUcTm9kZUpTIGZyYW1ld29ya3MsIEV4cHJlc3NKUw%3D%3D
21–24	Routing, static files, exceptions	CLO-3	Editor	Hands-on	https://expressjs.com/en/starter/static-files.html	https://www.youtube.com/watch?v=-wxw2-I2CtI&t=45s&pp=ygUhUm91dGluZywgc3RhdGljIGZpbGVzLCBleGNlcHRpb25z
25–28	Middleware & lifecycle	CLO-4	Editor	Demo	https://www.ibm.com/docs/en/was-nd/8.5.5?topic=configurations-adding-complete-life-cycle-middleware-servers	https://www.youtube.com/watch?v=1oWPUpMheGk&pp=ygUWTWlkZGxld2FyZSAmIGxpZmVjeWNsZQ%3D%3D
29–32	SSR vs CSR, EJS/HBS	CLO-5	Editor	Practical	https://www.geeksforgeeks.org/javascript/server-side-rendering-vs-client-side-rendering-vs-server-side-generation/	https://www.youtube.com/watch?v=CEWR1P5EpeY&t=200s&pp=ygUTU1NSIHZzIENTUiwgRUpTL0hCUw%3D%3D
33–36	SQL vs NoSQL, MongoDB, Mongoose	CLO-6	Editor	Hands-on	https://www.mongodb.com/resources/basics/databases/nosql-explained/nosql-vs-sql	https://www.youtube.com/watch?v=OdgZ0jr4jpM&pp=ygUsaW4gZW5nbGlzaCAtLVNRTCB2cyBOb1NRTCwgTW9uZ29EQiwgTW9uZ29vc2U%3D
37–40	Sessions, cookies	CLO-6	Editor	Demo	https://www.geeksforgeeks.org/javascript/difference-between-session-and-cookies/	https://www.youtube.com/watch?v=GhrvZ5nUWNg&t=326s&pp=ygUeaW4gZW5nbGlzaCAtLVNlc3Npb25zLCBjb29raWVz
41–44	Auth: bcrypt, JWT, passport	CLO-6	Editor	Practical	https://docs.nestjs.com/security/authentication	https://www.youtube.com/watch?v=CV1BryWmUdQ&pp=ygUoaW4gZW5nbGlzaCAtLUF1dGg6IGJjcnlwdCwgSldULCBwYXNzcG9ydA%3D%3D
45–48	Socket.io real-time apps	CLO-6	Editor	Hands-on	https://socket.io/get-started/chat	https://www.youtube.com/watch?v=hzf0EM1y4hY&list=PL63c_Ws9ecIRZ6njHRi3cuCkNSfzqyLBn

49–52	PostgreSQL, Prisma	CLO-6	Editor	Practical	https://www.prisma.io/postgres	https://www.youtube.com/watch?v=dFQ4f6RARCw&pp=ygUfaW4gZW5nbGlzaCAtLVBvc3RncmVTUUwsIFByaXNtYQ%3D%3D
53–56	Multer, Cloudinary	CLO-6	Editor	Demo	https://www.npmjs.com/package/multer-storage-cloudinary	https://www.youtube.com/watch?v=RNkAtObIrI4&pp=ygUfaW4gZW5nbGlzaCAtLU11bHRlciwgQ2xvdWRpbmFyeQ%3D%3D
57–60	Testing & deployment	CLO-6	Editor	Demo	https://www.geeksforgeeks.org/software-testing/post-deployment-testing-in-software-testing/	https://www.youtube.com/watch?v=J9JbzsufemE&pp=ygUhaW4gZW5nbGlzaCAtLVRlc3RpbmcgJiBkZXBsb3ltZW50

8. Remedial Classes[3]

After every Continuous Evaluation, different types of learners will be identified and special discussions will be planned and scheduled accordingly.

Action Plan for different types of learners:

Learner Type-I	Learner Type- II	Learner Type- III
Remedial Classes, Doubt Sessions, Guided Tutorials	Workshop, Doubt Session	Coding Competitions, Project

9. Self-Learning[4]

Assignments to promote self-learning, survey of contents from multiple sources.

S.No	Topics	CLO	ALM	References/MOOCS
1	Introduction to Client Server Architecture, Understanding Server and how request is handled at the server. Installing NodeJS and setup the environment, file handling module and diving into file dependency. Understanding Nodejs, advantages and disadvantages with respect to the available languages. Handling request, creating endpoints using HTTP module. Modules, NPM, import Modules. Frameworks, Different frameworks of NodeJS starting with Express-JS. Serve static files, Routing (Routing methods, Route paths, Route parameters, Route Handlers), Response Methods. Handling static pages with file stream, Handling exceptions.	CLO01-CLO03	Leading Question, Test Questions, and Practical Assignments	https://www.simplilearn.com/what-is-client-server-architecture-article https://www.geeksforgeeks.org/node-js-introduction/ https://expressjs.com/en/guide/routing.html https://expressjs.com/en/guide/routing.html https://expressjs.com/en/guide/routing.html
2	Middleware: Middleware lifecycle, Application-level, middleware, Router-level. Middleware, Error-handling middleware and Third-party middleware, how request travels in express, Blocking vs non-blocking code -Body parser. SSR vs CSR and Diving into Template Engines, how to use the EJS/HBS template in your Node.js application. Introduction to databases, SQL and No SQL Databases, Connecting MongoDB, Learning ODM-Mongoose with NodeJS application. Diving into session management and understanding Cookies and Express-Sessions. Diving into authentication in Express-JS with Bcrypt, and understanding how JWT token	CLO05-CLO6	Leading Question, Test Questions, Brain Storming, and Practical Assignments	https://expressjs.com/en/guide/using-middleware.html https://nodejs.org/en/learn/asynchronous-work/overview-of-blocking-vs-non-blocking https://www.geeksforgeeks.org/difference-between-sql-and-nosql/ https://www.geeksforgeeks.org/how-to-connect-node-js-to-a-mongodb-database/

[3] Refer to Annexure
[4] Refer to Annexure

	works, and passport JS. Deep Diving into full duplex communication and mastering socket.io			
3	Starting with PostgreSQL CRUD operation, and deep diving into PRISMA. Understanding Multer, Cloudinary to handle form-data. Learning unit testing and deployment on Vercel/Render/AWS etc.	CLO05-CLO6	Leading Question, Test Questions, Brain Storming, and Practical Assignments	https://docs.nestjs.com/security/authentication https://socket.io/get-started/chat https://www.prisma.io/postgres https://www.npmjs.com/package/multer-storage-cloudinary https://www.geeksforgeeks.org/software-testing/post-deployment-testing-in-software-testing/

10. Delivery Details of Content Beyond Syllabus[5]

Content beyond syllabus covered (if any) should be delivered to all students that would be planned, and schedule notified accordingly.

S.No	Advanced Topics, Additional Reading, Research papers and any	CLO	POs	ALM	References/MOOCS
1	Advanced NodeJS Internals: Event Loop, Asynchronous Programming, Performance Optimization	CLO-4	PO1, PO2, PO5	Self-learning, Demo, Case Study	https://nodejs.org/en/learn/asynchronous-work/event-loop https://www.digitalocean.com/community/tutorials/nodejs-performance-best-practices
2	Modern Backend Architectures: Microservices, REST vs GraphQL, API Documentation (Swagger/OpenAPI)	CLO-5	PO2, PO3, PO5, PO9	Project-based Learning, Comparative Study	https://www.geeksforgeeks.org/microservices-architecture-in-node-js/ https://graphql.org/learn/ https://swagger.io/docs/specification/about/
3	Secure and Scalable Backend Systems: OWASP Top 10, NoSQL Data Modeling, Database Design Patterns	CLO-6	PO2, PO3, PO6, PO8	Case Study, Design Exercise	https://owasp.org/www-project-top-ten/ https://www.mongodb.com/developer/products/mongodb/data-modeling-introduction/
4	Cloud-native Development & Deployment: CI/CD Pipelines, Docker, Cloud Deployment (AWS/Azure)	CLO-9	PO5, PO11, PO12	Workshop, Demonstration	https://docs.github.com/en/actions https://docs.docker.com/language/nodejs/ https://aws.amazon.com/getting-started/hands-on/deploy-nodejs-web-app/
5	Emerging Technologies & Research Trends: Real-time Applications (WebSockets), Serverless Architecture	CLO-7	PO2, PO4, PO5, PO12	Research Paper Review, MOOC-based Learning	https://ieeexplore.ieee.org/document/8451966 https://www.coursera.org/learn/serverless-cloud-computing

[5] Refer to Annexure

11. Evaluation Scheme & Components:

Assessment Type	Evaluation Component	Type of Component[6]	No. of Assessments	% Weightage of Component	Max. Marks	Mode of Assessment	CLO
Formative	Component 1	Project-based Evaluation I	01*	30%	30	Viva Voce, PPT Presentation, File work	CLO01- CLO02
Formative	Component 2	Project-based Evaluation II	01*	30%	30	Viva Voce, PPT Presentation, File work	CLO01- CLO04
Summative	Component 3	Final Project-based Evaluation	01**	40%	40	**Project**, Viva Voce, PPT Presentation, File work	CLO01- CLO06
	Total		**100%**				

* Students will have to appear in all the Project-based evaluations.
** As per Academic Guidelines, a minimum 75% attendance is required to become eligible for appearing in the Final Project-based Evaluation.

12. Syllabus of the Course:

Subject:			
S.No.	**Topic(s)**	**No. of Lectures**	**Weightage %**
1	Introduction to Client Server Architecture, Understanding Server and how request is handled at the server. Installing NodeJS and setup the environment, file handling module and diving into file dependency. Understanding Nodejs, advantages and disadvantages with respect to the available languages. Handling request, creating endpoints using HTTP module. Modules, NPM, import Modules. Frameworks, Different frameworks of NodeJS starting with Express-JS. Serve static files, Routing (Routing methods, Route paths, Route parameters, Route Handlers), Response Methods. Handling static pages with file stream, Handling exceptions.	24	40%
2	Middleware: Middleware lifecycle, Application-level, middleware, Router-level. Middleware, Error-handling middleware and Third-party middleware, how request travels in express, Blocking vs non-blocking code -Body parser. SSR vs CSR and Diving into Template Engines, how to use the EJS/HBS template in your Node.js application. Introduction to databases, SQL and No SQL Databases, Connecting MongoDB, Learning ODM-Mongoose with NodeJS application. Diving into session management and understanding Cookies and Express-Sessions. Diving into authentication in Express-JS with Bcrypt, and understanding how JWT token works, and passport JS. Deep Diving into full duplex communication and mastering socket.io	48	80%
3	Starting with PostgreSQL CRUD operation, and deep diving into PRISMA. Understanding Multer, Cloudinary to handle form-data. Learning unit testing and deployment on Vercel/Render/AWS etc.	60	100%

13. Academic Integrity Policy:

Education at Chitkara University builds on the principle that excellence requires freedom where Honesty and integrity are its prerequisites. Academic honesty in the advancement of knowledge requires that all students and Faculty respect the integrity of one another's work and recognize the importance of acknowledging and safeguarding intellectual property. Any breach of the same will be tantamount to severe academic penalties.

This Document is approved by:

Designation	**Name**	**Signature**
Course Coordinator	Dr. Kiran Deep Singh	
Head-Academic Delivery	Dr. Mrinal Paliwal	

Dean	Dr. Rishu Chhabra	
Date(DD/MM/YYYY)	15/01/2026	

Annexure

1. **Pre- requisite**
   Mention The Pre-requisite skill set or course/s if it is expected to be studies before this course, otherwise write "not applicable".

2. **NHEQF levels**

The NHEQF levels represent a series of sequential stages expressed in terms of a range of learning outcomes against which typical qualifications are positioned/located. NHEQF level 4.5 represents learning outcomes appropriate to the first year (first two semesters) of the undergraduate programme of study, while Level 8 represents learning outcomes appropriate to the doctoral-level programme of study.

Table 1: Higher education qualifications at different levels on the NHEQF

NHEQF level	Examples of higher education qualifications located within each level
Level 4.5	Undergraduate Certificate. Programme duration: First year (first two semesters) of theundergraduate programme, followed by an exit 4-credit skills-enhancement course(s).
Level 5	Undergraduate Diploma. Programme duration: First two years (first four semesters) of the undergraduate programme, followed by an exit 4-credit skills-enhancement course(s) lasting two months.
Level 5.5	Bachelor's Degree. Programme duration: First three years (Six semesters) of the four-yearundergraduate programme.
Level 6	Bachelor's Degree (Honours/ Honours with Research). Programme duration: Four years(eight semesters).
Level 6	Post-Graduate Diploma. Programme duration: One year (two semesters) for those whoexit after successful completion of the first year (two semesters) of the 2-year master's programme.
Level 6.5	Master's degree. (e.g. M.A., M.Com., M.Sc., etc.) Programme duration: Two years (four semesters) after obtaining a 3- year Bachelor's degree (e.g. B.A., B.Sc., B.Com.etc.).
Level 6.5	Master's degree. (e.g. M.A., M.Com., M.Sc., etc.) Programme duration: One year (two semesters) after obtaining a 4 - year Bachelor's degree (Honours/ Honours with Research) (e.g. B.A., B.Sc., B.Com. etc.).
Level 7	Master's degree.(e.g. M.E./M.Tech. etc.) Programme duration: Two years (four semesters) after obtaining a 4-year Bachelor's degree. (e.g. B.E./B.Tech. etc.)
Level 8	Doctoral Degree

3. **NHEQF level descriptors**

Each NHEQF level is structured based on the defined learning outcomes which lead to the expected graduate attributes/profile. The level descriptors reflect the expected outcomes of learning that should be achieved and demonstrated by graduates of a specific programme of study leading to a qualification at a specific NHEQF level.

Click Learning outcomes descriptors for qualification for all levels on the NHEQF

4. **Course Outcomes**
   The number of Course Outcomes is recommended to be 4-5 for courses that do not contain practical component and 6 for those courses with a practical component. Flexibility can be sought by the post-graduate courses in this regard.

5. **Theory/lab Plan**
   The following are the guidelines to be followed while creating plans

- Each session may be planned for a duration of 45/50mins (irrespective of the double hour or single hour scheduled in timetable).
- Every session must incorporate at least one active learning method which may or may not be part of the assessments.
- Put BoS Approved Syllabus in the topics. Deviations (if any) from BoS approved syllabus must be brought to the notice of BoS chairman & Dean Academics, After approval, revised handout should be submitted.
- The Topics elaborated in the Theory/Lab plan must match those in the course execution plan.

6. **Teaching Learning Methods**
   The following are some of the Teaching & Learning methods that can be incorporated in session wise teaching learning plan.
   - **Teacher-centered Learning Methods:**
     i. Lecture
     ii. Discussion
     iii. Demonstration method using a simulation or a tool

iv. Reviewing
v. Questioning

- **Learner-centered teaching & Learning methods:**
  i. **Active learning**, in which students solve problems, answer questions, formulate questions of their own, discuss, explain, debate, or brainstorm during class;
  ii. **Cooperative learning**, in which students work in teams on problems and projects under conditions that assure both positive interdependence and individual accountability; and
  iii. **Inductive teaching and learning**, in which students are first presented with challenges (questions or problems) and learn the course material in the context of addressing the challenges.
  iv. **Inductive methods** include inquiry-based learning, case based instruction, problem-based learning, project-based learning, discovery learning, and just-in-time teaching. It is important to integrate authentic, reflective and collaborative learning experiences when designing for student-centered learning.

7. **Active Learning Methods**

The following are some of the Active Learning Methods that can be incorporated in session wise teaching learning plan.

- One Minute Paper
- Group Discussion
- Student-Created PPT, Charts, Matrices, Flowcharts, Models
- The Fish Bowl
- Debate
- Video Synthesis
- Quiz/Test Questions
- Brain Storming Sessions
- Case Study
- Shadowing
- Leading Question
- Puzzle, Enigma, Contradiction
- Statement-Opinion-Summary
- Think / Pair / Share
- Peer Review
- Just in Time Teaching
- Statement-Opinion-Summary
- Peer Survey
- Focused Listing
- Role-Playing
- Student Field Work with Reflection
- Infusing Humor into Class Sessions
- Inviting Effective Guest Speakers

8. Remedial Classes

After every Sessional Test, identify weak learners, provide supplement course handout. Student list and Impact Observed report should be submitted to Dean through proper channel.

9. Self Learning

Plan 10% of topics in self-learning mode with discussions, ALM's and Assessment happing in the class.

10. Content Beyond Syllabus

Plan Advanced Topics, Experiments, Additional Reading, Research papers in self-learning mode with ALM's and Assessment happing in the regular class or lab. Usually caters advanced learners. Identify Advanced learners. For Extra classes, schedule should be notified accordingly.

11. Assessment Type

1. Assessment broadly can be classified into the following types:

a. **Diagnostic assessments**: Diagnostic assessments are intended to help teachers identify what students know and can do in different domains to support their students' learning. These help teachers determine strengths of students in various areas to
better address their specific needs.
b. **Formative assessments**: Formative assessment refers to a wide variety of methods that teachers use to conduct in-process evaluations of student comprehension, learning needs, and academic progress during a lesson, unit, or a course. Formative assessments help teachers identify concepts that students are struggling to understand, skills they are having difficulty acquiring, or learning standards they have not yet achieved so that adjustments can be made to lessons, instructional techniques, and academic support.
c. **Summative assessments**: Summative assessment is an assessment administered at the end of an instructional unit in a course. These assessments are intended to evaluate student learning by comparing performance to a standard or benchmark.
d. **Ipsative assessments**: Ipsative assessment involves comparisons between past and current work to identify a learner's growth over time, rather than progress toward an external set of criteria. Therefore, Ipsative assessment is an internal or self-referenced assessment.
e. **Norm-referenced assessments**: Norm-referenced tests report whether test takers performed better or worse than a hypothetical average student, which is determined by comparing scores against the performance results of a statistically selected group of test takers, typically of the same age or grade level, who have already taken the exam.
f. **Criterion-referenced assessments**: Criterion-Reference tests measure the performance of test takers against the criteria covered in the curriculum.
g. **Peer-to-Peer randomised Assessments**: Peers will be able to provide assessment in this case
h. **Industry Validation of Effectiveness**: In the Vocation Education, Industry validation of effectiveness of training is particularly important.
i. **Self-assessments**: To evaluate how much the learner has grasped by self-learning.

2. Other Assessment Methods: Conducting an assessment takes time, thought, attention, planning, and often collaboration. Each assessment tool, whether a short survey or detailed rubric, will be useful only insofar as it both addresses the outcomes well and is feasible to use.

a**. Rubrics**: For assessing qualitative student work such as essays, projects, reports, or presentations. Rubrics serve well to clearly denote the specific expectations for an assignment, for collecting data for assessment of student learning outcomes. and for student performance. Rubrics can be used for grading, for providing feedback to students, and for informing and encouraging students to think about their own learning.
b. **Portfolios and E-Portfolio**: Portfolios can provide a window into the process of student learning across a semester-long project that can be assessed (usually by using a rubric).
c. **Curriculum Mapping**: A good curriculum map can serve to focus assessment, and the improvements that follow, where it will be most useful, informative, or effective.
d. **Structured Interviews**: While time-consuming, structured interviews are useful when specific questions need to be asked. It also leaves room for unplanned topics or ideas to emerge.
e. **Student Experience Surveys**: Student experience in research universities (SERU), including administration of on-line census SERU Undergraduate and Graduate Surveys, can yield important information about student perceptions and experiences.

12. Evaluation Component & Types

As per LMs we need to figure it out whether it is component 1, 2 or 3. In Types of Evaluation Component, we need to specify what type of evaluation we are performing like Continuous Evaluation or Sessional Test or End Term Examination.

13. No. of Assessments and Weightage of Components

Department will give guideline for number of assessments, mandatory or optional and weightage.